



Course Outline

Catalog Information:

CS 56 Networking Essentials 3.000 Units

This course is a general introduction to current network technologies for local area networks (LANs), wide area networks (WANs), and the Internet. The course includes an introduction to the hardware, software, terminology, components, design, and connections of a network, as well as the topologies and protocols for LANs. This course includes LAN-user concepts and the basic functions of system administrations and operation. (CSU)

Requisites:**Advisories**

CS 11 with a minimum grade of C

Authorized Discipline(s):

- Computer Information Systems
- Computer Science

TOP Code (CB03)

0708.00 - Computer Infrastructure and Support*

SAM Code (CB09)

D - Possibly Occupational

Type	Units	In-Class Hours	Outside-of-Class Hours	Total Student learning Hours
Lecture	3.00	54.00	0.00	54.00
Laboratory	0.00	0.00	0.00	0.00
Total	3.00	54.00	0.00	54.00

TBA Lab Hrs: Min

TBA Lab Hrs: Max

Course Repeatability: No

Course Objectives:

Upon successful completion of this course, the student will be able to:

1. Identify the components of a LAN and determine the type of network design most appropriate for a given site.
2. Identify the different media used in network communications, distinguish between them, and determine how to use them to connect servers and workstations in a network.
3. Distinguish between the different networking standards, protocols, and access methods, and determine which would be most appropriate for a given LAN.
4. Recognize the primary network architectures, identify their major characteristics, and determine which would be the most appropriate for a proposed LAN.
5. Identify the primary functions of network operating systems and distinguish between a centralized computing environment and a client/server environment.
6. Determine how to implement and support the major networking components (including the server, operating system, and clients), and propose a system for adequately securing data on a given LAN and protecting the system's components.
7. Distinguish between LANs and WANs, identify the components used to expand a LAN into a WAN, and determine how to implement an appropriate modem in the larger LAN/WAN environment.

8. Identify strategic LAN support tools and resources, and determine how to use these in troubleshooting basic network problems.
9. Identify the components of the Internet.

Course Learning Outcomes:

Upon successful completion of this course, the student will be able to:

1. Students will demonstrate an understanding of basic networking design.
2. Students will demonstrate understanding of basic networking diagnostics

Outline of Topics:

1. Network Orientation:
 1. Introduction to Networks
 2. Network Configuration
 3. Network Topology
2. Network Components:
 1. Network Cabling
 2. Network Interface Card
 3. Networking Devices; repeater, bridge, hub, switch and router.
 4. Wireless Networking
 5. WAN technology; PSTN, X.25, Frame Relay, PSDN, T1 Carriers, DSL and ATM
3. Network Architectures:
 1. Access Methods
 2. How Networks Send Data
 3. Ethernet
 4. Token Ring
4. Network Operating System:
 1. Introduction to Network Operating Systems
 2. Microsoft Desktop Operating Systems (Windows 7 or 8)
 3. Microsoft Network Operating Systems (Windows Server)
 4. Unix or Linux Network Operating Systems
5. Network Standards:
 1. Open Systems Interconnection (OSI) Reference Model
 2. The IEEE 802.x Standard
6. Network Protocols
 1. All protocols in the OSI reference model
 2. IPV4 addressing
 3. Classfull and Classless subnetting
 4. Introduction to to IPV6
 5. Address Translation
7. Virtual Network and Remote Access
 1. Virtualization
 2. Virtual Network Components
 3. VLAN
 4. Cloud Computing
 5. Remote Access
 6. Virtual Private Network (VPN)
8. Network Security
 1. Security Risks
 2. Security in Network Design
 3. Encryption
 4. Authentication Protocols
 5. Wireless Network Security
9. Network Administration and Support:

1. Designing and Installing a Network
2. Establishing Network Shares and Accounts
3. Ensuring Network Security
4. Printing on a Network
5. Administering Change
6. Troubleshooting a Network

Sample Assignments:

Reading:

Students may be asked to complete research, which requires use of the Internet, library resources, or written articles, on topics such as:

1. What is the difference between packet switching and circuit switching?
2. What are the differences between how these network devices work: hub, switch, router and a firewall?
3. Explain how a star topology network differs from that of a bus topology network.

Writing:

Writing assignments can be short research papers or expository writing assignments which allow the students to delve more deeply into a particular area of interest. As an example, the instructor might provide students with essential features of how a protocol such as http works in the application layer, and then ask the student to do a detailed investigation of all the protocols in the TCP/IP model and explain clearly how each protocol works.

Out-of-Class:

Students may review magazine articles or websites on the importance of security in http (https) because they are the ones we use in our daily online communication.

Methods of Instruction:

Methods of instruction may include, but are not limited to, the following: (from methods of instruction data page).

- Lecture

Distance Education

Will any part of this class be taught online?

Yes

Choose ONE of the following options:

Choose which part of the course will be online:

Note: When choosing an emergency only component you are agreeing to: DE-only instruction of class component will only be implemented if a government-declared or district-declared emergency makes in-person instruction infeasible or impossible.

Lecture

Lab

Accessibility

All course materials must be accessible to students with disabilities. Title 5 requires that distance education in the California Community Colleges is subject to the requirements of the federal Americans with Disabilities Act and section 508 of the Rehabilitation Act of 1973.

What steps will be taken to ensure course content and assignments are ADA compliant? (select all that apply) The choices here represent the basic actions to complete that will help make your course accessible to students with disabilities. It is recommended to choose all of them.

Other No

Professor-Initiated Interaction

Regular Substantive Interaction (RSI) between the instructor and students is mandated in Title 5 for all Distance Education courses, regardless of whether the course is fully online or delivered as a hybrid. In the case of a hybrid, regular substantive interaction - initiated by the instructor-must occur in the online portion of the class. **Title 5 states that courses taught online must have a minimum of two of the following five interactions types present in the course. Please indicate which instructor-initiated interactions will be present in this course **Must select a minimum of 2:****

Student-to-Student Interaction

Regular interaction among students is mandated in Title 5. This is necessary to design a collaborative, student-centered environment in which a community of learners is created. How will course outcomes and regular contact among students, either synchronously or asynchronously, be achieved in this course?

Check all that apply:

Other No

Methods of Evaluation:

A student's grade will be based on a required final examination and multiple measures of performance including critical thinking. These methods may include, but are not limited to the following:

- Essay Examinations
- Objective Examinations
- Problem-Solving Examinations
- Reports
- Essays
- Research Papers
- Projects
- Case Study Analysis
- Classroom Participation
- Quizzes
- Homework Assignments
- Classroom Assignments
- Lab Assignments
- Oral Presentations

Other Method of Evaluation

1) Other evaluation methods such as computer assignments, computer programs.

Textbooks:

1. Mike Meyers. *CompTIA Network+ Certification All-in-One Exam Guide* 7th. McGraw-Hill. 2019.
2. Ramon Nastase. *Computer Networking for Beginners* 6th or current edition. Ramon Nastase. 2018.
3. Beasley & Nilkaew. *Networking Essentials* 5th. Pearson. 2018.

Manuals: None

Periodicals: None

Other Instructional Materials:

Comparable Courses:

1. **Community College Course**
Las Positas College
Networking Fundamentals
CNT52
2. **CSU**
CSU Monterrey Bay
Intro to Computer Networks
CST 311

Board of Trustees Approval Date:

02/04/2020